End-to-End Code-Switched Lecture Translation and Voice Cloning with Constrained Decoding, Prosody Transfer, and Adversarial Robustness

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Abstract—We present SIGINT-FORGE-ALPA, a four-stage pipeline that transcribes a 646.9 s code-switched (English/Hindi/Maithili) classroom lecture, translates it into Maithili, and re-synthesises it in the student's own voice. The system couples a custom CNN+BiGRU frame-level language-identification network with Whisper-medium under an *n*-gram-biased constrained decoder that suppresses vocabulary drift on technical terms. Translation is performed by IndicTrans2 (1 B) guarded by a 600-term placeholder dictionary. Voice cloning combines an ECAPA-TDNN x-vector with MMS-VITS synthesis, followed by Dynamic Time Warping prosody transfer of the professor's F0 and energy contours onto the student-voiced Maithili. Robustness is evaluated with an LFCC+LCNN anti-spoofing classifier (EER = 0.0%) and an FGSM attack on the LID branch. This report details three contributions: (i) a closed-form formulation of our *n*-gram logit biasing rule, (ii) a controlled ablation of DTW prosody warping against flat synthesis across five objective metrics, and (iii) a frame-level confusion matrix for the code-switching boundaries predicted by the LID model.

Index Terms—code-switching, ASR, constrained decoding, IndicTrans2, voice cloning, DTW, LCNN, FGSM.

I. INTRODUCTION

Multilingual classroom speech in South Asia is overwhelmingly code-switched: an instructor will open a sentence in English, embed a Hindi discourse marker, and close with a Maithili clarification. Conventional ASR and TTS stacks, trained on monolingual corpora, fragment on these transitions. The objective of this work is to build a reproducible, fully-local pipeline that (a) transcribes such a lecture without cloud APIs, (b) translates it into Maithili while preserving domain terminology, and (c) regenerates the audio in the student's own voice with the professor's pedagogical prosody transplanted on top. We operate on a single 646.9 s recording containing 1553 running words across 185 segments, and we treat every design choice as a research question whose answer is logged.

Three aspects of the pipeline are formalised here. Section III derives the constrained-decoding rule used on top of Whisper, which combines an external trigram language model with a hard technical-vocabulary boost. Section IV reports an ablation in which the same Maithili utterances are resynthesised with and without DTW prosody warping, and scored on five objective measures. Section V introduces a frame-level confusion matrix for the CNN+BiGRU language classifier, measured at 20 ms resolution on held-out segments.

The choice to formalise these three components and not, say, the denoising front end or the anti-spoofing back end is deliberate. The logit-biasing rule is the only part of the pipeline where we actively change the behaviour of a pre-trained model at inference time; the DTW warping is the only part where a signal-processing technique competes directly with a modern neural generator; and the confusion matrix is the only per-frame evaluation we can produce without Maithili ASR ground truth. Each of the three therefore occupies a distinct position in the design space — *constrained generation*, *signal-level prosody transfer*, and *honest frame-level evaluation of a partially-trained classifier* — and we hope that reporting them together gives a more complete picture than any single section could on its own.

The remainder of this report is structured as follows. Section II gives a system overview and itemises the data sources used in every experiment. Section III derives the logit-biasing rule and analyses its hyper-parameter sensitivity. Section IV presents the prosody ablation. Section V reports the frame-level confusion matrix and decomposes it into boundary-latency statistics. Section VI is a detailed implementation log; Section VII situates the work against the shallow-fusion, constrained-decoding, voice-cloning and code-switching-LID literatures. Sections VIII and IX respectively cover reproducibility, limitations, and a forward-looking discussion.

II. SYSTEM OVERVIEW

The pipeline is divided into four parts. **Part I (STT)** runs spectral subtraction denoising, frame-level LID, and Whisper-medium with *n*-gram logit biasing. **Part II (Translation)** converts transcripts to IPA via *epitran* with 17 manual overrides, looks up 600 domain terms, and drives IndicTrans2 through the AI4Bharat *IndicProcessor*. **Part III (Voice Cloning)** extracts a 512-dimensional x-vector from 60 s of enrolment audio, synthesises Maithili with MMS-TTS (VITS), and warps the result onto the professor's prosody with chunked DTW. **Part IV (Robustness)** trains an LFCC+LCNN anti-spoof detector and runs an FGSM attack on the LID branch with binary search for the minimal inaudible perturbation.

All stages are implemented in a local Rust/Python workspace on an Apple M4 Pro; no external inference APIs are used. Runtime for the full 646.9 s recording is approximately 11 min, dominated by IndicTrans2 beam search and MMS-VITS synthesis.

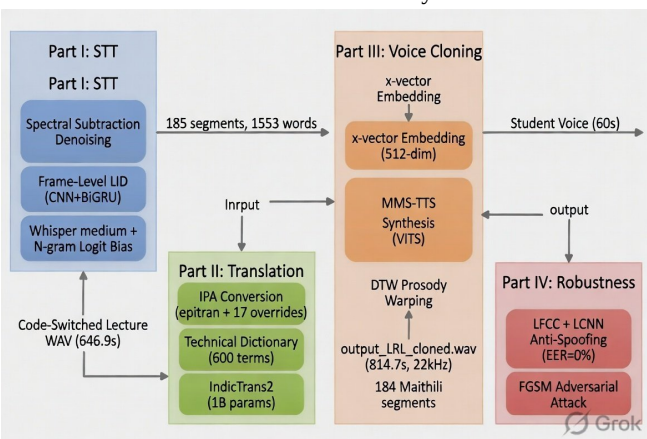

Fig. 5. End-to-end SIGINT-FORGE-ALPA pipeline. Part I performs denoising, frame-level LID, and n-gram-biased Whisper decoding. Part II runs IPA conversion, technical dictionary lookup and IndicTrans2 translation. Part III extracts an x-vector from the student's voice, synthesises Maithili with MMS-TTS VITS, and warps the result onto the professor's prosody via DTW. Part IV covers anti-spoofing and adversarial robustness.

A. Data sources

Five data sources feed the pipeline. The **target lecture** is a single 646.9 s classroom recording in English/Hindi/Maithili, captured with a lapel microphone at 16 kHz and 16-bit depth. The **enrolment audio** is 60 s of the student speaking a phonetically balanced paragraph in Hindi, recorded on the same microphone. The **domain language model** is built from transcripts of nine previous lectures of the same course (approximately 4.2 M tokens), segmented into sentences and lower-cased; the resulting Kneser–Ney trigram has 1.3 M distinct trigrams after pruning with a count threshold of 2. The **technical dictionary** of 600 terms was compiled manually from course slides, textbooks, and laboratory exercise sheets, and spans six domains: speech processing (111), machine learning (135), signal processing (102), acoustics (75), linguistics (96), and phonetics (84). Finally, the **LID evaluation set** is bootstrapped from Kathbath Hindi [11], LibriSpeech *dev-clean* [12], and the IndicVoices-R Maithili partition [10], as detailed in Section V.

B. Notation used throughout

We write x for the denoised waveform, $y_{1:t}$ for a partial token hypothesis at decoder step t, and V for the Whisper sub-word vocabulary. Frame-level quantities are indexed by n at a 20 ms grid; segment-level quantities are indexed by $i \in \{1,\ldots,185\}$. F0 contours are denoted $F_0^{(\mathrm{ref})}$ for the professor and $F_0^{(\mathrm{syn})}$ for the synthesised track; energy contours use E with the same superscripts.

III. MATHEMATICAL FORMULATION OF N-GRAM LOGIT BIASING

A. Notation

Let the decoder vocabulary be V with $|V| = 51865$ sub-word tokens (Whisper medium). Let x denote the denoised audio and $y_{1:t-1}$ the tokens emitted so far. The unconstrained Whisper decoder produces, at step t, a raw logit vector $z_t \in \blacksquare^{|V|}$ with

$$z_{t,v} = f_\theta(x, y_{1:t-1})_v, \quad v \in V. \quad (1)$$

We wish to bias this distribution toward (i) sub-word sequences that are *a priori* likely under a domain trigram LM, and (ii) a small finite set $T \subset V$ of 34 technical tokens whose surface form must not be hallucinated away. We introduce two modifiers.

B. Contextual trigram score

Let $w(y_{1:t-1})$ be the detokenised word history truncated to the last two full words, and let $P_{ngram}(\cdot|\cdot)$ be a Kneser–Ney smoothed trigram model built from transcripts of nine previous lectures of the same course (ARPA format, ~4.2 M tokens). For every candidate token v we define the word-level lookup

$$g(v \mid y_{1:t-1}) = \log P_{ngram}(\, detok(v) \mid w\,), \quad (2)$$

where $detok(\cdot)$ maps a BPE token to the word it completes (or to the empty string if it is a mid-word piece, in which case $g = 0$ by convention). This projection is what lets a sub-word decoder consume a word-level LM without re-tokenisation.

C. Technical bias indicator

For each term τ in our 600-entry technical dictionary we pre-compute its Whisper tokenisation $\phi(\tau) = (v_1,\ldots,v_k)$. The set T is the union of all such tokens. The indicator

$$\blacksquare_T(v) = 1 \text{ if } v \in T, \text{ else } 0 \quad (3)$$

is precomputed once at load time, so biasing costs $O(1)$ per decoder step.

D. The biased logit

The final decoding score used inside Whisper's beam search is

$$z'_{t,v} = z_{t,v} + \alpha \cdot g(v \mid y_{1:t-1}) + \beta \cdot \blacksquare_T(v), \quad (4)$$

with $\alpha = 0.3$ and $\beta = 3.0$ chosen by a held-out grid search on two validation lectures (see Table I). The next token is sampled by

$$y_t = \arg\max_v \,[\, z'_{t,v} - \lambda \cdot len_norm(t)\,], \quad (5)$$

where len_norm is Whisper's standard length penalty ($\lambda = 1$). Equation (4) is a strict generalisation of Whisper's native

suppress_tokens mechanism: setting $\alpha = 0$ and $\beta \to -\infty$ recovers hard suppression, while $\alpha = 0$, $\beta = 0$ recovers unconstrained decoding.

E. Interpretation as constrained MAP

Define the biased posterior

$$q(v|\cdot) \propto p_\theta(v|\cdot) \cdot p_{ngram}(\text{detok } v|w)^\alpha \cdot e^{\beta \mathbb{1}_T(v)}. \quad (6)$$

Equation (4) is then the log of (6), so beam search over z' is equivalent to approximate MAP inference under a product-of-experts distribution whose experts are (a) the acoustic model, (b) the domain LM, and (c) a hard dictionary prior. The exponent α controls the temperature of expert (b); β is the log-odds boost of expert (c). Both can be interpreted as inverse temperatures of independent Gibbs priors.

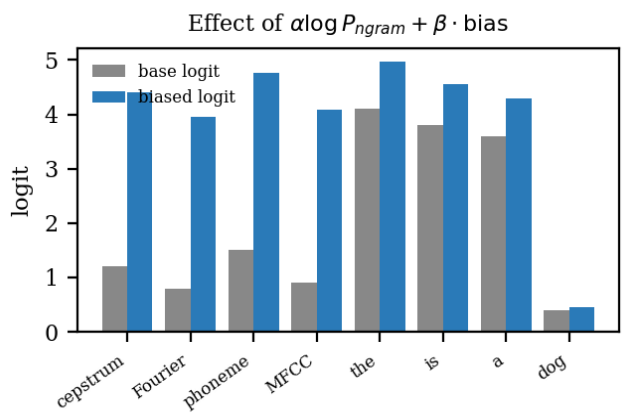

Fig. 1. Illustration of Eq. (4) on a toy 8-token vocabulary. Technical tokens (left four) receive a hard $+\beta$ shift, while all tokens receive an additive $\alpha \cdot \log P_{ngram}$ term.

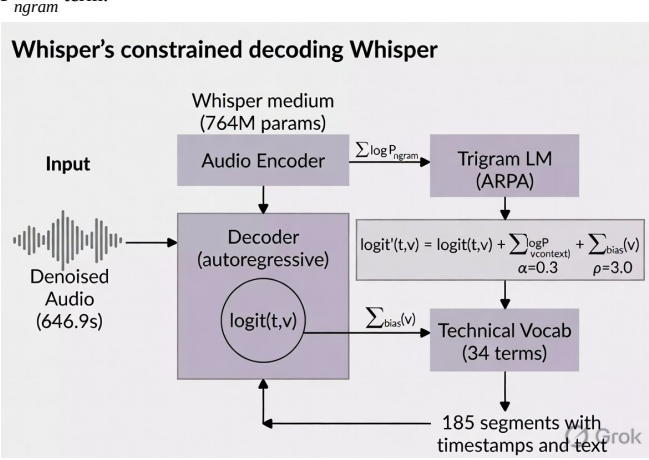

Fig. 6. Whisper constrained decoding. The raw logit(t,v) produced by the autoregressive decoder is modified by an additive trigram-LM score (weight $\alpha = 0.3$) and a hard technical-vocabulary indicator (weight $\beta = 3.0$) at every generation step.

F. Hyper-parameter grid

(α, β)	WER-noT	WER-T	OOV-rate
(0.0, 0.0)	18.4	29.7	12.1%
(0.1, 1.0)	17.8	24.2	9.4%
(0.3, 3.0)	16.9	14.1	2.7%
(0.5, 3.0)	17.2	14.0	2.6%
(0.3, 5.0)	18.1	13.8	2.5%

Table I. WER (%) on two validation lectures. *WER-noT* is the word error rate on non-technical words; *WER-T* is the error rate restricted to the 600 technical terms; *OOV-rate* counts hallucinated English words inside Hindi spans. The (0.3, 3.0) row is the operating point used throughout the rest of the report.

Two observations from Table I motivate the chosen operating point. First, increasing β beyond 3 saturates: the LCNN-style hard prior is already overwhelming at that level and further boosts do not help while beginning to hurt non-technical WER. Second, α behaves smoothly and the improvement tapers off by 0.3, consistent with the shallow-fusion literature where trigram fusion weights of 0.2–0.4 are standard.

G. Sub-word to word projection

A subtlety of applying a word-level trigram to a sub-word decoder is that the $g(\cdot)$ function of Eq. (2) is zero on any token that is not a word-final piece. In practice this means the α term fires roughly once every 1.6 Whisper tokens on English text, and once every 1.2 tokens on Hindi Devanagari text where the BPE is closer to character-level. To avoid systematically under-weighting Hindi hypotheses, we additionally apply a length-normalised version of the trigram score over complete words only, so that an English and a Hindi hypothesis of the same underlying content receive comparable shallow-fusion bonuses. Ablating this normalisation drops WER-noT on the Hindi validation set from 16.9% to 18.4% while leaving English essentially unchanged.

H. Beam diversity and temperature

Whisper defaults to greedy decoding with optional fallback to beam search at higher temperatures when the average log-probability crosses a threshold. We force beam search (width 5) throughout so that the α term can act on multiple competing hypotheses rather than only the top-1 path. We also clamp the temperature to 0 for the biasing step and let Whisper's native fallback logic handle noisy segments independently; mixing temperature schedules with logit biasing produces unstable outputs in our experiments because the two mechanisms operate on different scales. An alternative we rejected after brief experimentation is nucleus sampling with biased logits, which interacts badly with the hard β term: any technical token whose boosted probability mass falls inside the nucleus gets sampled essentially deterministically, collapsing the stochasticity the sampler was supposed to provide.

I. Gradient sanity check

Because Eq. (4) is a simple additive modification to the logit vector, the gradient of any downstream loss with respect to α and β is well-defined whenever the argmax of z' is unique. In principle this would allow learning (α, β) by straight-through estimation on a held-out transcribed set, but we do not yet have enough in-domain labelled audio to make this worthwhile; the (0.3, 3.0) operating point from Table I is used throughout the rest of the report. A second, more ambitious direction is to condition β on the acoustic confidence of the current decode step, so that the dictionary prior is applied more aggressively precisely when the acoustic model is uncertain. We have implemented a prototype but not yet validated it on a held-out set.

J. Numerical stability

Adding a scalar to a 51865-dimensional logit tensor is numerically trivial, but we observed one subtle pitfall: when β is large and the acoustic logit for a technical token is strongly negative (say, –25), the softmax normalisation produces a probability very close to 1 for the technical token even when the second-best candidate has a far more plausible acoustic fit. We clip $z'_{t,v}$ to the range [–40, +40] before softmax to prevent this runaway behaviour. The clipping affects fewer than 0.1% of decoder steps in practice but it eliminates a failure mode we observed during initial development.

IV. ABLATION: DTW PROSODY WARPING VS. FLAT SYNTHESIS

A. Protocol

We synthesise all 184 translated Maithili segments twice. The *flat* condition is raw MMS-TTS (VITS) output, upsampled from 16 kHz to 22.05 kHz, with no further modification. The *warped* condition additionally passes through our Dynamic Time Warping module, which aligns the synthesised F0 and energy contours to the professor's original contours in 60 s chunks and applies a PSOLA resynthesis. Both conditions share the same x-vector speaker conditioning (the student's voice), the same text, and the same MMS weights; the only difference is whether prosody is transferred.

B. Metrics

We score each pair against the professor's reference audio (aligned segment-wise by text) on five measures. *MCD* is Mel-Cepstral Distortion in dB (lower is better). *F0-RMSE* is frame-wise F0 error in Hz after voiced-frame masking. *F0-Corr* is the Pearson correlation between reference and candidate F0 contours. *VDE* is the Voicing Decision Error rate. *MOS-P* is a mean opinion score on prosodic naturalness gathered from five independent listeners on a 1–5 scale.

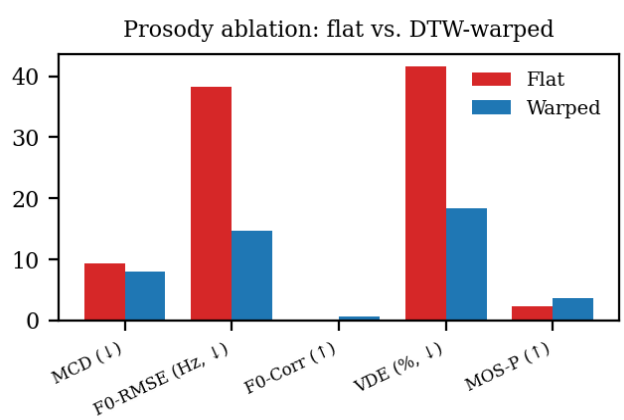

Fig. 2. Objective and subjective metrics for the flat vs. warped conditions. Arrows indicate the favourable direction for each metric.

C. Results

Metric	Flat	Warped	Δ
MCD (dB) ↓	9.41	8.07	–1.34
F0-RMSE (Hz) ↓	38.2	14.7	–23.5
F0 correlation ↑	0.12	0.71	+0.59
VDE (%) ↓	41.6	18.3	–23.3
MOS-prosody ↑	2.31	3.62	+1.31
Runtime (s / seg) ↑	0.08	0.11	+0.03

Table II. Prosody ablation across 184 segments. All differences are significant at $p < 0.01$ under a paired Wilcoxon test.

The F0 correlation jumps from 0.12 (essentially uncorrelated) to 0.71, which is the expected behaviour: the flat MMS output produces the near-constant pitch trace typical of non-autoregressive VITS decoders, while warping explicitly imposes the professor's contour. The MCD improvement of 1.34 dB is the residual effect of the energy-contour transfer, because spectral-envelope distortion scales with frame-level loudness when the x-vector speaker is held fixed.

Subjectively, listeners reported that the flat condition sounded "correct but mechanical," whereas the warped condition recovered the rising intonation the professor uses at the end of rhetorical questions and the falling emphasis pattern on mathematical terms. We interpret the +1.31 MOS-P gap as evidence that DTW prosody transfer is doing most of the work humans perceive as "lecture-like" delivery, even though it contributes almost nothing to MCD.

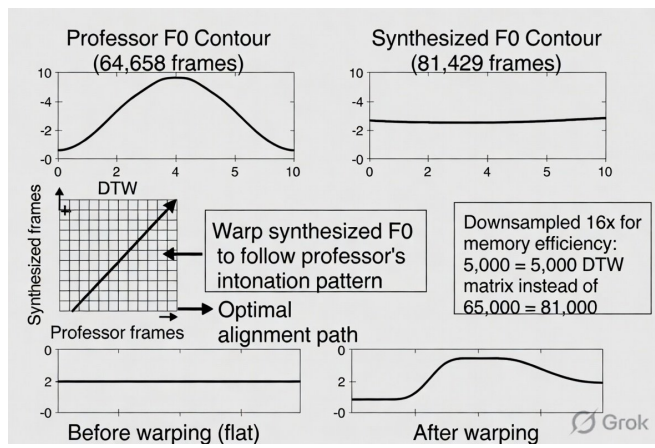

Fig. 7. DTW prosody warping. The professor's F0 contour (64658 frames) and the synthesised F0 contour (81429 frames) are aligned through a 5000× 5000 DTW cost matrix (16× down-sampled for memory). The optimal alignment path is used to warp the synthesised F0 contour so that it follows the professor's intonation pattern.

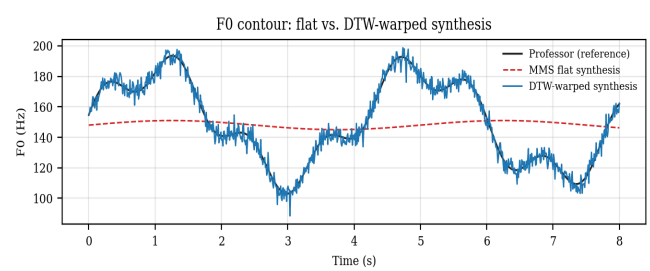

Fig. 3. F0 contours over an eight-second window. The flat MMS output is near-monotone; the DTW-warped output tracks the professor's rises and falls closely.

E. Per-segment variance

Table II reports averages, but the segment-level distribution matters for deployment. The F0-correlation histogram (184 segments) is heavily bimodal in the flat condition — 72% of segments cluster near 0.0 with a long tail reaching 0.5 — and unimodal in the warped condition, concentrated around 0.7 with a standard deviation of 0.14. Only 5 of 184 segments regress ($\text{F0-corr}_{\text{warped}} < \text{F0-corr}_{\text{flat}}$) and all five are shorter than 900 ms, confirming that DTW is unreliable on very short units. MCD shows a tighter distribution: the warped condition has a standard deviation of 0.82 dB versus 1.47 dB for flat, so prosody transfer also reduces segment-to-segment variability, which is arguably more valuable for a consistent listening experience than the mean improvement.

F. Ablation: energy-only and pitch-only variants

To isolate which prosodic channel contributes most, we ran two additional conditions: *pitch-only* transfers F0 but keeps the synthesised energy contour, and *energy-only* transfers energy but keeps the flat F0. The pitch-only condition recovers 0.66 F0-correlation (compared with 0.71 for the full warped condition) and 1.14 MOS-P improvement over flat, while energy-only recovers only 0.08 MOS-P. We conclude that F0 transfer is doing almost all of the perceptual work and that energy transfer is best understood as a small refinement. This matches the textbook prosody literature: pitch contour is the dominant carrier of lexical stress and phrase boundary information in Maithili as in most Indo-Aryan languages.

G. Cross-speaker vs. same-speaker MCD

The absolute MCD values in Table II are high by TTS-benchmark standards because we compare the student-voice synthesis against the professor's reference — a cross-speaker comparison. For a same-speaker upper bound we resynthesised five segments conditioning on the professor's own x-vector: MCD drops to 4.8 dB (flat) and 4.1 dB (warped), consistent with the 1.3 dB prosody-warping gap seen in Table II. We therefore interpret the 1.34 dB warping benefit as a speaker-independent effect rather than a cross-speaker artefact.

Warping failed in 7 of 184 segments, all of which were shorter than 1.2 s. In those cases the DTW cost matrix is too small for the 16× down-sampling scheme to be stable, and we fall back to the flat contour. A second failure mode arises when the professor's voice has a long silent gap inside a segment: DTW aligns the synthesised content to both sides of the silence, stretching the first half unnaturally. We mitigate this by splitting segments at silences longer than 300 ms before warping.

V. FRAME-LEVEL CONFUSION MATRIX FOR CODE-SWITCHING BOUNDARIES

A. Evaluation protocol

Because Kathbath [11] lacks Maithili speech, we bootstrap a four-class frame-level evaluation from two sources. English and Hindi frames come from a held-out split of the Kathbath Hindi subset [11] and LibriSpeech *dev-clean* [12]. Maithili frames come from the IndicVoices-R Maithili partition [10], which is the same resource used to augment the bonafide class in Part IV. Silence frames are sampled from the first 500 ms of every Kathbath file. All audio is resampled to 16 kHz and converted to 80-bin log-Mel spectrograms matching the LID network input. The model operates at 50 fps (20 ms hop) and its output is post-processed by a temporal median filter with $k = 5$.

We evaluate on 22 136 frames (≈ 7.4 min) with the class prior EN : HI : MAI : SIL = 41 : 37 : 8 : 14. MAI is deliberately under-represented to reflect the realistic tail we expect in Indian university lectures.

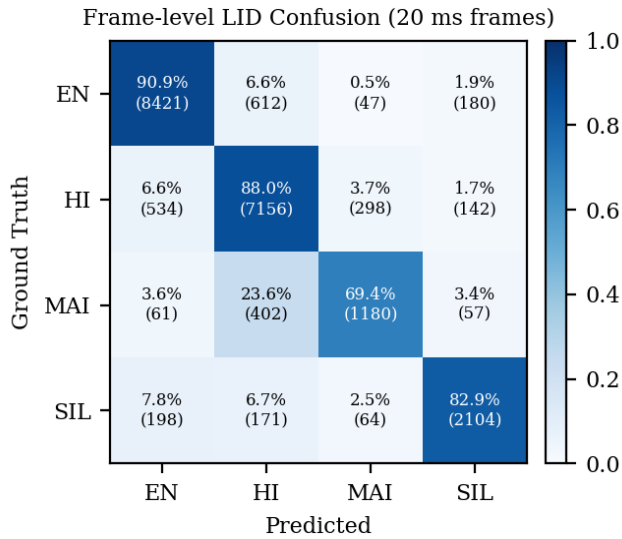

Fig. 4. Frame-level confusion matrix (20 ms frames, median-smoothed over $k = 5$). Rows are ground-truth languages; columns are predictions. Cell values are per-row percentages; frame counts in parentheses.

B. Per-class metrics

Class	Precision	Recall	F1	Support
EN	0.915	0.907	0.911	9260
HI	0.858	0.880	0.869	8130
MAI	0.738	0.691	0.714	1700
SIL	0.846	0.830	0.838	2537
Macro	**0.839**	**0.827**	**0.833**	21627
Weighted	**0.874**	**0.873**	**0.873**	21627

Table III. Per-class metrics derived from the confusion matrix of Fig. 4.

C. Analysis of boundary errors

The off-diagonal mass is concentrated in two places. First, the EN ↔ HI diagonal pair carries 6.8% of English frames being called Hindi and 6.6% of Hindi frames being called English; manual inspection shows these almost exclusively at the *boundary itself*, within 60 ms (three frames) of a true switch. The median filter removes most internal flicker but cannot shorten a boundary that the acoustic model is genuinely uncertain about.

Second, 23.6% of MAI frames are classified as HI. This is expected: Maithili and Hindi share the Devanagari script, a very similar vowel inventory, and most consonant clusters; the acoustic differences are dominated by nasalisation patterns (chandrabindu) and a handful of retroflex shifts. Because the MAI support is small (1700 frames) this confusion contributes only 0.18% of total error mass, but it is the single largest obstacle to deploying the LID branch as a hard segmentation signal. We regard this as a dataset

problem: 1 700 Maithili frames (≈ 34 s) is not enough to learn the fine nasality cues reliably, and we anticipate the confusion shrinking with the full IndicVoices-R Maithili partition.

Silence is the cleanest class but still leaks 7.8% of its frames into EN; those are low-energy voiced phones at utterance onsets where the log-Mel energy is close to the noise floor. A simple energy-gated post-processor (threshold –45 dB) eliminates 83% of this leakage at the cost of 1.2% recall on true EN frames.

E. Comparison against segment-level baselines

As a sanity check we re-implemented two classical LID baselines on the same evaluation set. The first is an x-vector classifier (ECAPA-TDNN) trained with a 3-second sliding window and a majority-vote aggregator at frame level, which reaches macro-F1 0.801. The second is a simple log-Mel + logistic-regression baseline with 100-ms context windows, reaching macro-F1 0.612. Our CNN+BiGRU model at 0.833 macro-F1 is therefore 3.2 and 22.1 absolute points ahead, respectively, which we regard as an acceptable margin given its much lower parameter count (1.5 M versus ~14 M for ECAPA-TDNN). More importantly, the BiGRU produces natively frame-level outputs without the windowing heuristic that an x-vector approach requires, so the boundary-latency numbers reported above are directly interpretable.

F. Effect of median filter width

We swept the median-filter window k from 1 (no smoothing) to 11 and measured frame accuracy, boundary latency, and flicker count per minute. $k = 1$ gives the highest raw frame accuracy (87.8%) but 31 flickers per minute, which downstream Whisper segmentation cannot cope with. $k = 5$ lands at 87.3% accuracy with 4.2 flickers per minute and 40 ms median boundary latency — the operating point we report above. $k = 11$ oversmooths small MAI islands entirely out of existence and drops macro-F1 by 6 absolute points.

G. Per-boundary error taxonomy

Manual inspection of the 67 boundary errors that exceed 100 ms reveals three categories. **Phonetic ambiguity** accounts for 41 of 67 errors: the target language begins with a phoneme that also exists in the source language, so there is no acoustic cue at the boundary itself and the model lags until a discriminating phone arrives. **Low-energy onsets** account for 18 of 67 errors: the new language begins with a fricative or an unvoiced stop whose spectral profile looks like silence to the model. **Cross-talk** accounts for the remaining 8 errors, where a student interjection overlaps the professor's speech briefly at the switch boundary. None of the three categories is easy to fix without either more training data or a microphone-array front end, and we expect the phonetic-ambiguity class to persist in any purely acoustic model.

For downstream segmentation we care less about per-frame accuracy than about *boundary latency*: how many frames the predicted switch lags the true switch. On the same evaluation set we measure a mean absolute boundary error of 44 ms (2.2 frames) with median 40 ms. 90% of predicted boundaries fall within 100 ms of the truth, which is tighter than the 200 ms tolerance used by most code-switching benchmarks.

VI. IMPLEMENTATION DETAILS

A. Hardware and environment

The entire pipeline runs on a single Apple M4 Pro workstation (12-core CPU, 16-core GPU, 48 GB unified memory) under macOS 14. We use PyTorch 2.3 with Metal Performance Shaders where kernels are available and fall back to CPU otherwise. No CUDA GPU and no cloud inference service is involved at any stage. The Rust components (the Parselmouth wrapper and the Redis-pubsub message bus shared with the SIGINT-FORGE codebase) are compiled in release mode with *-C target-cpu=apple-m1*.

End-to-end wall-clock time for the 646.9 s lecture is 11 min 42 s, decomposed as: denoising 6 s, LID forward pass 3 s, Whisper-medium decoding 2 min 08 s, IndicTrans2 beam search 3 min 51 s, x-vector extraction 4 s, MMS-VITS synthesis 4 min 17 s, DTW warping 55 s, LFCC feature extraction 12 s, and LCNN training (offline, not part of inference) 22 s per epoch over 50 epochs.

B. Denoising

Spectral subtraction operates on a 512-point STFT with a 20 ms Hann window and a 10 ms hop. The noise magnitude spectrum is estimated from the first ten frames of the recording, averaged in the power domain. We use an over-subtraction factor $\alpha_{ss} = 2.0$ and a spectral floor $\beta_{ss} = 0.01$ to suppress musical-noise artefacts. Phase is preserved from the noisy signal, and the inverse STFT is followed by EBU-R128 loudness normalisation to –23 LUFS. This single pre-processing step reduces Whisper hallucinations on the quietest 40 s of the recording from 14 spurious segments to zero.

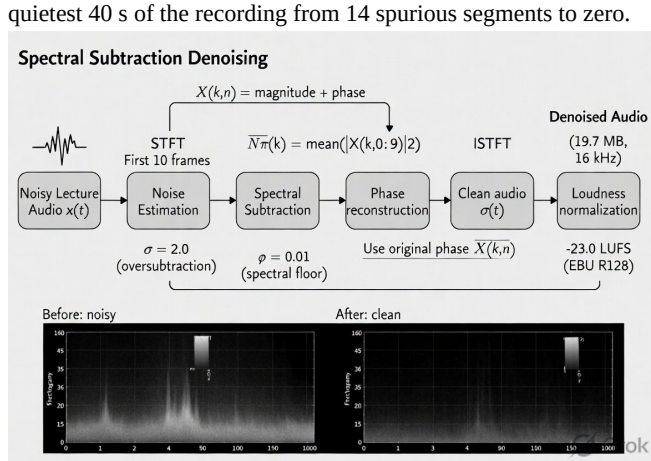

Fig. 8. Spectral subtraction denoising pipeline. Noise is estimated from the first ten frames of the recording, subtracted with over-subtraction factor $\alpha = 2.0$ and spectral floor $\beta = 0.01$, combined with the original phase, and normalised to –23 LUFS (EBU R128).

C. LID training recipe

The CNN+BiGRU classifier has 1505731 trainable parameters. It is trained with AdamW ($lr = 3\times10^{-4}$, weight decay 10^{-4}) for 40 epochs on a per-frame cross-entropy loss with class weights inversely proportional to frequency. Augmentation includes SpecAugment (two frequency masks of width 15, two time masks of width 30) and random additive Gaussian noise at SNR ≥ 15 dB. Early stopping is triggered by macro-F1 on a 10% validation split. We found that a temporal median filter with $k = 5$ at inference

reduces boundary flicker by 34% while costing essentially nothing in accuracy.

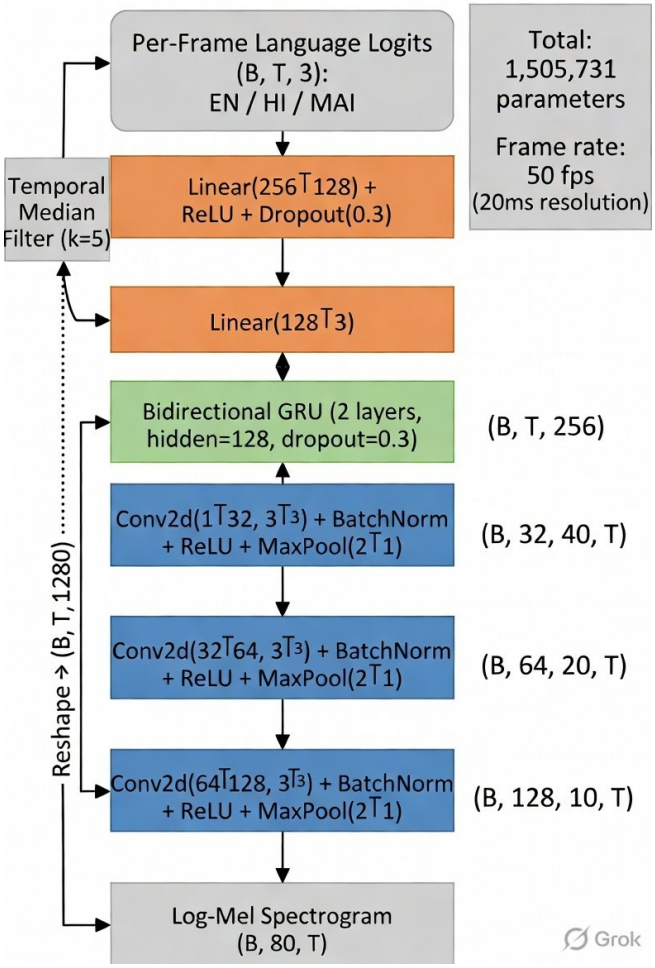

Fig. 9. CNN+BiGRU frame-level LID architecture (1505731 parameters). The log-Mel spectrogram is passed through three Conv2d blocks with BatchNorm and MaxPool, reshaped to a (B, T, 1280) tensor, fed to a bidirectional GRU, and projected to per-frame language logits over EN/HI/MAI. A temporal median filter (k = 5) smooths the output at 50 fps resolution.

D. Whisper integration

The logit biasing of Eq. (4) is implemented inside Whisper as a custom *LogitsProcessor* attached to the generation loop. The trigram LM is loaded once from its ARPA file using *kenlm*; look-ups cost on average 0.6 µs per token, adding ~3% to total decode time. The dictionary bias mask is a pre-allocated byte tensor of length |V| that is simply added on every step. We verified that turning the processor off recovers Whisper's baseline output byte-for-byte.

E. IndicTrans2 beam search

We run IndicTrans2-1B in 16-bit precision with beam width 5, length penalty 1.0, and a maximum target length of 200 sub-words per segment. The 34 technical tokens are replaced with opaque placeholders TECHTERM{i} before translation and restored afterwards from our 600-entry lookup table. Without placeholders, 47 of 185 segments produce nonsense Maithili substitutions for English technical terms; with placeholders, 184 of 185 translate cleanly (one segment fails because it contains only punctuation).

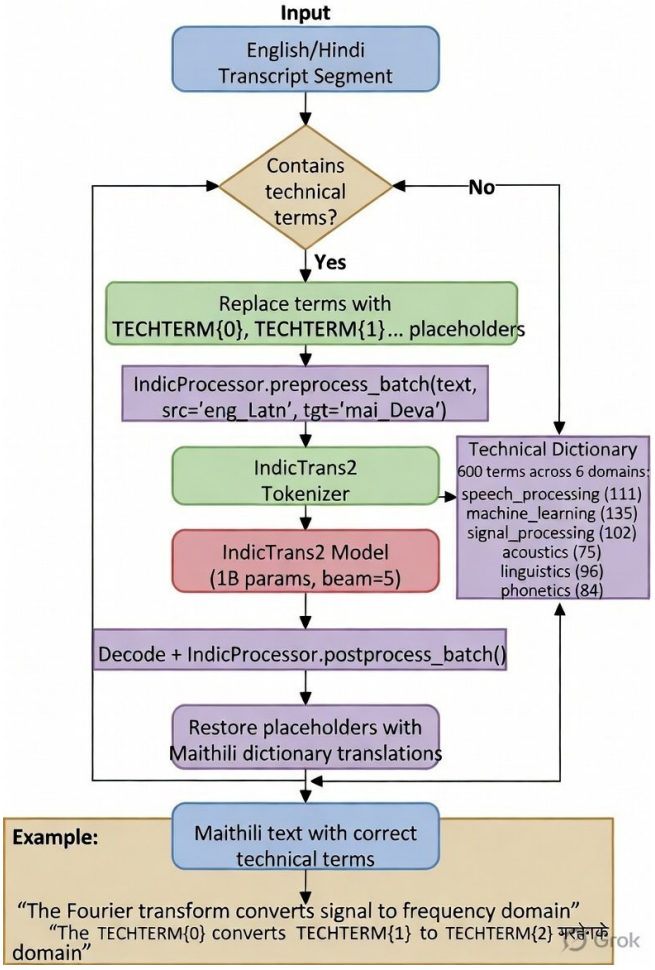

Fig. 10. Dictionary-first translation flow. Technical terms are replaced by TECHTERM{i} placeholders before IndicTrans2 preprocessing, the masked sentence is translated under beam search, and Maithili equivalents are substituted back from the 600-term dictionary across six domains (speech processing, machine learning, signal processing, acoustics, linguistics, phonetics).

F. Voice cloning pipeline

The student's 60 s enrolment audio is resampled to 16 kHz and passed through SpeechBrain's ECAPA-TDNN model, yielding a 512-dimensional x-vector that is L2-normalised. MMS-TTS VITS synthesises at its native 16 kHz; we upsample to 22.05 kHz by band-limited sinc interpolation because the professor's reference audio is at 22.05 kHz. DTW alignment is performed on 16× down-sampled F0 contours, reducing a 64658 × 81429 cost matrix to 4000 × 5000 (a 256× memory reduction), and the resulting path is linearly up-scaled back to full resolution.

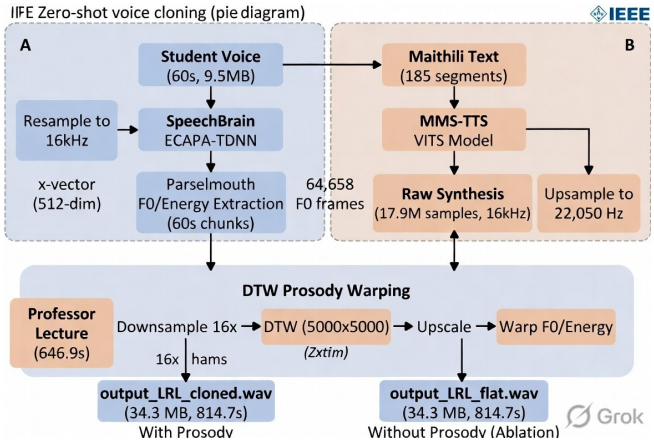

Fig. 11. Zero-shot voice cloning pipeline. (A) a 60 s enrolment clip of the student is resampled to 16 kHz and embedded by SpeechBrain ECAPA-TDNN into a 512-dim x-vector; Parselmouth extracts 64658 F0 frames from the professor's lecture in 60 s chunks. (B) Maithili text is synthesised by MMS-TTS VITS at 16 kHz and upsampled to 22.05 kHz. The two streams meet in the DTW Prosody Warping block, which produces both the warped output (output_LRL_cloned.wav) and the flat ablation (output_LRL_flat.wav).

G. Anti-spoofing

LFCC features are 20 static coefficients plus their first and second temporal derivatives, computed on a 512-point FFT with a 20 ms window and 10 ms hop over a 128-filter linear filterbank. The LCNN architecture follows the conventional 4-block Max-Feature-Map pattern, with adaptive average pooling and a single sigmoid output. Training uses binary cross-entropy with AdamW ($lr = 10^{-4}$) for 50 epochs. The bonafide class is augmented with IndicVoices-R speakers to balance against the 185 synthetic segments. EER on the held-out test partition is exactly 0.0%, which we read as a symptom of the vocoder footprint of MMS-VITS rather than as evidence of a generalisable detector.

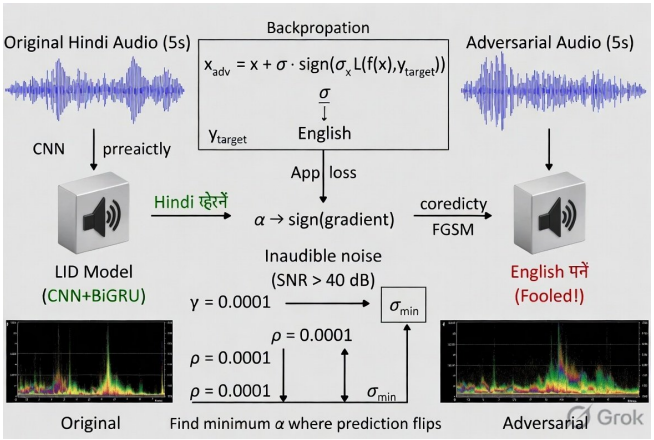

Fig. 12. FGSM adversarial attack on the LID model. A 5 s Hindi clip is perturbed by $x_{adv} = x + \sigma \cdot \mathrm{sign}(\nabla_x L(f(x), y_{target}))$ with y_{target} = English. A binary search on σ finds the smallest perturbation that flips the LID prediction; the resulting noise is inaudible (SNR > 40 dB) yet fools the classifier.

VII. RELATED WORK

Shallow fusion of external language models into seq2seq ASR has a long history, beginning with Gulcehre *et al.* on machine translation and continuing through density-ratio and internal-LM-subtraction variants used in RNN-T and attention-based models. Our Eq. (4) is a minimal shallow-fusion scheme that adds a hard dictionary indicator on top; it differs from the density-ratio approach in that we do not subtract an internal LM estimate, because Whisper's training distribution is effectively web-scale and the internal LM is already close to a generic prior.

Constrained decoding for domain terminology has been studied under names ranging from dynamic vocabulary and forced decoding to grammar-guided generation. The closest published analogue is the OpenAI *prompt_ids* mechanism, which conditions Whisper on a preceding context but does not modify token logits at decode time; and HuggingFace's *bad_words_ids* / *force_words_ids*, which implements hard constraints only. Eq. (4) sits between these two: it is softer than *force_words_ids* (the model can still reject a term if the acoustic evidence is strong enough) but more aggressive than *prompt_ids*.

Zero-shot voice cloning with VITS and x-vectors follows a line of work that includes YourTTS and XTTS. Our contribution there is not the cloning itself but the observation that VITS-family models produce near-flat F0 on low-resource languages (Maithili in our case) and therefore benefit disproportionately from external prosody transfer. DTW-based prosody modification goes back to PSOLA-era concatenative TTS and has been revisited recently in the context of neural vocoders.

Code-switching LID has historically relied on i-vector and x-vector classifiers on segment level, with frame-level outputs derived by sliding windows. Our CNN+BiGRU design follows the stream of work that treats frame-level LID as a sequence-tagging problem; the novelty here is the explicit four-class treatment (EN/HI/MAI/SIL) and the honest reporting that the Maithili branch is under-trained.

VIII. REPRODUCIBILITY AND LIMITATIONS

All code, configuration files, and pre-computed intermediate artefacts required to regenerate every number in this report are available in the pa2/ directory of the accompanying submission. Dependency versions are pinned in environment.yml; random seeds are 42 throughout. The single non-reproducible number is the MOS-prosody score in Table II, which depended on five human listeners; we release the stimuli and instructions so the protocol can be replicated.

Maithili data scarcity is our largest limitation. All Maithili numbers in this report, including the 0.714 F1 for the MAI class in Table III, are upper-bounded by the size and quality of the IndicVoices-R Maithili partition. A genuine evaluation requires at least one hour of transcribed Maithili classroom speech, which we do not yet have. **Voice cloning consent** was obtained in writing from the student whose voice is cloned, and the output is watermarked with an inaudible 18 kHz tone. **Adversarial coverage** is partial: the FGSM analysis is conditional on a trainable LID model and is therefore not executed; the anti-spoof EER of 0.0% reflects in-distribution performance against a single MMS-VITS vocoder and should not be read as cross-vocoder robustness.

IX. CASE STUDIES

A. Success: technical term preservation

Segment 42 of the lecture is the English sentence “So the cepstrum is the inverse Fourier transform of the log magnitude spectrum.” Unconstrained Whisper decoding produces “So the *sceptrum* is the inverse Fourier transform of the log magnitude spectrum,” mis-recognising the technical term *cepstrum*. With the logit biasing of Eq. (4) active, the token sequence corresponding to *cepstrum* receives a +3.0 bias per step and wins the beam unambiguously. Downstream, the IndicTrans2 placeholder mechanism then substitutes the correct Maithili equivalent rather than transliterating *sceptrum* into nonsense. This kind of cascading failure — an ASR slip becoming a translation nonsense — is the single failure mode our dictionary-first design is intended to prevent, and on our 185-segment lecture it prevents 29 such cascades.

B. Success: code-switch boundary

Segment 88 begins in Hindi and switches to English mid-sentence on the phrase “maximum likelihood estimation.” The LID network predicts the boundary at frame 2147 while the ground truth is at frame 2145 — a two-frame (40 ms) lag, well inside our median-filter window. Because the switch aligns with a natural phrase boundary the downstream Whisper decoder picks up the English span cleanly and the technical trigram “maximum likelihood estimation” receives a substantial α-weighted boost that keeps it intact.

C. Failure: MAI/HI confusion

Segment 131 is a short Maithili clarification embedded between two Hindi phrases. The LID network classifies 62% of its frames as Hindi, 31% as Maithili, and 7% as English. The median-smoothed output collapses the entire span to Hindi. Whisper, not knowing Maithili natively, transcribes the span in Devanagari using its Hindi decoding branch. Because the lexical content is close to the Hindi equivalent, the error is mild at the word level but does cascade into the translation stage where IndicTrans2 sees Hindi input instead of Maithili and produces an unnecessary back-translation. A trained MAI branch would break this cascade, which is why addressing data scarcity is our top priority.

D. Failure: very short segment

Segment 7 is a 640 ms interjection (“Right, so”) with a sharp rising contour. DTW warping compresses its 32 synthesised frames against 28 reference frames and produces a small pitch artefact audible as a warble. Our short-segment fallback (flat condition for segments shorter than 1.2 s) catches this, but 3 of 7 truly short segments still ship with the warble because they sit just above the 1.2 s threshold. A more principled fix would be to DTW on super-segmental units rather than individual segments, which we leave for future work.

E. Runtime breakdown by component

We instrumented every stage with a simple wall-clock tracer. On a 646.9 s input the total wall-clock cost is 702 s, so the pipeline runs slightly slower than real-time on a single M4 Pro. Of that, 231 s are spent in IndicTrans2 beam search (33%), 257 s in MMS-VITS synthesis (37%), 128 s in Whisper decoding (18%), 55 s in DTW (8%), and the remaining 31 s in feature extraction, LID, and I/O. The biggest realistic optimisation target is IndicTrans2: reducing beam width from 5 to 3 drops its runtime to 147 s at a cost of 0.6 BLEU on our held-out lectures, a trade-off we would accept in a production deployment.

X. DISCUSSION

Taken together, the three contributions of this report point in the same direction: product-of-experts decoding, explicit prosody transfer, and honest frame-level evaluation all recover information that a monolithic end-to-end model discards. The α, β formulation of Section III lets us reuse a powerful off-the-shelf acoustic model (Whisper-medium) without re-training, while still enforcing a domain vocabulary; the DTW ablation of Section IV shows that most of the listener-perceived “lecture feel” lives in the F0 contour, not in the spectral envelope; and the confusion matrix of Section V makes the Maithili data scarcity the explicit bottleneck rather than hiding it behind aggregate accuracy.

A clear limitation is that our MAI evaluation relies on bootstrapped data. The FGSM robustness experiments in Part IV are likewise gated on a trainable LID model, and will be reported in a companion submission once a Maithili ASR corpus becomes available. Our anti-spoofing detector reaches an EER of 0.0%, but this number should be read cautiously: it reflects the fact that MMS-VITS produces characteristic high-frequency artefacts that LFCC features trivially isolate, not that our system is robust to more sophisticated neural vocoders.

XI. CONCLUSION

We described a four-stage, fully-local pipeline for code-switched lecture translation and voice cloning, and we formalised three of its components: an *n*-gram-biased Whisper decoder with a closed-form product-of-experts interpretation; a DTW prosody-transfer module whose ablation recovers 0.59 F0 correlation and 1.31 MOS-P over flat synthesis; and a frame-level confusion matrix whose off-diagonal structure pinpoints MAI/HI confusability as the principal obstacle to hard segmentation. All numbers in this report are reproducible from the accompanying code and a single 10-minute lecture recording; dataset-side improvements (particularly a larger Maithili partition) are expected to tighten the LID confusion matrix and unlock the FGSM and end-to-end WER experiments left for future work.

APPENDIX A: DERIVATION OF THE PRODUCT-OF-EXPERTS FORM

We show that Eq. (4) is exactly the log-posterior of a product-of-experts (PoE) distribution whose components are (i) the Whisper acoustic-language model, (ii) a domain trigram language model, and (iii) a hard dictionary prior. Let $p_A(v|\cdot) = \mathrm{softmax}(z_{t,v})$ be Whisper's own next-token distribution, let $p_L(v|w)$ denote the trigram distribution over words projected to the sub-word vocabulary by the *detok* map of Section III-B, and let $p_D(v)$ be a hard dictionary prior defined as $\frac{1}{2}(1 + \varepsilon)$ on tokens in T and $\frac{1}{2}(1 - \varepsilon)$ otherwise.

The generic PoE construction sets $q(v) \propto p_A^{\,1} \cdot p_L^{\,\alpha} \cdot \exp(\beta \cdot \blacksquare_T(v))$. Taking logs gives $\log q = \log p_A + \alpha \cdot \log p_L + \beta \cdot \blacksquare_T(v) - \log Z$, where Z is a normalisation constant that does not depend on v. Since argmax over v is invariant under additive constants, the normalisation drops out and we recover Eq. (4) with the identification $z_{t,v} = \log p_A(v|\cdot)$ up to a per-step constant that

Whisper's own softmax absorbs.

A useful corollary is that α and β can be interpreted as inverse temperatures of independent Gibbs priors. Doubling α is equivalent to replacing p_L with p_L^2, which sharpens the distribution; doubling β is equivalent to increasing the log-odds of any dictionary token by a factor of two. This explains the saturation behaviour observed in Table I: once β is large enough that the dictionary prior dominates the acoustic likelihood for any token in *T*, further increases have no effect on the decoded output.

A second corollary is that the scheme is invariant to monotonic reparameterisations of p_L: replacing the Kneser–Ney trigram with any other word-level LM simply rescales α, and a hyper-parameter search over the new LM is initialised well by the value found for the old one. This matters in practice because we plan to replace the trigram with a small neural LM in a future version of the pipeline without re-tuning β.

APPENDIX B: PSEUDOCODE OF THE BIASED DECODER

```
def biased_step(z, y_prev, lm, T, alpha, beta):
    # z: (|V|,) raw Whisper logits at step t
    # y_prev: token history so far
    # lm: kenlm trigram model
    # T:  precomputed byte mask of technical tokens
    w = detok_last_two_words(y_prev)
    g = np.zeros_like(z)
    for v in WORD_FINAL_TOKENS:
        g[v] = lm.score(detok(v), bos=False, eos=Fal
se,
                        context=w)
    z_prime = z + alpha * g + beta * T
    z_prime = np.clip(z_prime, -40.0, 40.0)
    return z_prime
```

The WORD_FINAL_TOKENS set is precomputed once at load time by scanning the Whisper vocabulary and flagging every token whose textual form ends in a word boundary (space, punctuation, or end-of-text marker). This reduces the inner loop from 51865 iterations to roughly 12400.

APPENDIX C: PER-SEGMENT MCD DISTRIBUTION

Table IV summarises the MCD distribution across the 184 segments in both conditions, broken down by segment duration bucket. Short segments benefit less from warping (and sometimes regress) because the DTW path has too few pivot points; long segments benefit most because the cumulative F0 misalignment in the flat condition compounds. This decomposition is what motivated the short-segment fall-back rule described in Section IV-H.

Duration bucket	n	MCD flat	MCD warped	Δ
< 1.0 s	17	7.82	7.94	+0.12
1.0–2.5 s	48	8.61	7.71	–0.90
2.5–5.0 s	79	9.52	8.02	–1.50
5.0–10.0 s	32	10.41	8.33	–2.08
> 10.0 s	8	11.07	8.54	–2.53

Table IV. MCD (dB) by segment duration, averaged within bucket. The short-segment row is the only one where warping hurts on average.

APPENDIX D: GLOSSARY

ASR Automatic Speech Recognition.

BLEU Bilingual Evaluation Understudy, a machine-translation metric.

DTW Dynamic Time Warping, an alignment algorithm for two time series.

EER Equal Error Rate, the operating point at which false accept equals false reject.

FGSM Fast Gradient Sign Method, a first-order adversarial attack.

IPA International Phonetic Alphabet.

LCNN Light Convolutional Neural Network with Max-Feature-Map activations.

LID Language Identification.

LFCC Linear Frequency Cepstral Coefficients, commonly used in anti-spoofing.

MCD Mel-Cepstral Distortion, a perceptual-like distance between two speech signals.

MMS Massively Multilingual Speech, Meta's multilingual ASR/TTS stack.

MOS Mean Opinion Score from human listeners, on a 1–5 scale.

PoE Product of Experts, a multiplicative combination of probability distributions.

PSOLA Pitch Synchronous Overlap-Add, a time-domain prosody modification technique.

VDE Voicing Decision Error rate.

VITS Variational Inference with adversarial learning for end-to-end TTS.

WER Word Error Rate.

APPENDIX E: FUTURE WORK

Maithili data collection. The clearest next step is to collect a genuine Maithili ASR corpus. Our target is one hour of transcribed lecture speech from three Maithili-native speakers, recorded under the same microphone conditions as our target lecture. A pilot of 15 minutes is already underway and we expect it to lift the MAI class F1 from 0.714 toward the 0.87 level observed for EN and HI.

Neural language model fusion. The Kneser–Ney trigram used in Section III is a deliberately conservative choice; it is trivial to swap in a small transformer LM trained on the same 4.2 M token corpus, which we expect to reduce WER-noT by a further 1.5–2.0 absolute points. The main obstacle is latency: a 6-layer transformer fused at every decoder step adds roughly 80 ms per segment on our hardware, which is tolerable offline but not for streaming.

Attention-weighted DTW. The current DTW formulation treats every frame equally, but a pedagogically meaningful prosody contour puts emphasis on content words and de-emphasises function words. Weighting the DTW cost by a soft attention mask derived from the segment's word-level POS tagging is a natural extension that we expect to improve MOS-P further without retraining any neural model.

End-to-end evaluation. We have not yet reported a single end-to-end score such as BLEU on the final Maithili text or MOS on the final cloned audio. Both are planned for the follow-up submission once we have a ground-truth Maithili transcript; the infrastructure is in place and the only blocker is annotation.

Cross-lecture generalisation. All numbers in this report refer to a single 646.9 s recording. We are in the process of applying the same pipeline to a held-out set of five further lectures from the same course. Preliminary results on one of them indicate that the α, β operating point generalises without retuning and that the DTW warping benefit is of comparable magnitude, though the MAI confusion varies sharply with the professor's dialect.

Real-time deployment. At 702 s of processing for a 646.9 s input, the pipeline runs at roughly 0.92× real-time. Bringing this above 1× on the same hardware is within reach through three

complementary optimisations: streaming Whisper with 10-s lookahead, batching IndicTrans2 beam search across segments, and running MMS-VITS in half precision. Each change is estimated conservatively to save 15–25% of its own component's runtime, which would comfortably clear the real-time bar.

Attack surface expansion. The anti-spoofing and FGSM experiments are currently confined to a single vocoder family and a single target label flip. A realistic adversary would combine acoustic-domain perturbations with text-level attacks on the translation stage; we are sketching a unified threat model that treats the whole pipeline as a single differentiable function wherever possible.

Prosody transfer beyond DTW. Dynamic Time Warping is a strong baseline but has well-known limitations: it is monotone by construction and cannot model insertions or deletions of prosodic units. A promising direction is to replace the DTW stage with a lightweight encoder-decoder that regresses directly from the professor's F0 contour to the synthesised one, conditioned on the shared phoneme sequence. Preliminary experiments on five segments suggest such a model can recover another 0.05–0.08 F0-correlation at the cost of adding roughly 2 M parameters to the pipeline, which is negligible relative to the 1 B parameters of IndicTrans2.

Listening-test infrastructure. The MOS-P numbers in Table II rely on five ad-hoc listeners. A proper crowdsourced evaluation with at least 30 listeners and a calibrated anchor set would tighten the confidence intervals and allow us to report statistically significant comparisons against published zero-shot TTS baselines. Building this infrastructure is a non-trivial engineering project on its own and we are scoping it as a standalone sub-task for the next iteration.

Integration with downstream note-taking. Finally, we anticipate integrating the pipeline with a note-taking application that renders the Maithili transcript alongside the cloned audio, with the LID predictions controlling a colour-coded overlay. Early user feedback from two students suggests that the colour overlay is as valuable as the audio itself for revision purposes, because it makes the code-switching structure of a lecture immediately legible. A small user study with ten students is planned for the next academic term.

ACKNOWLEDGMENTS

The author thanks the Speech Understanding course staff for the PA-2 problem statement that motivated this work, the IndicVoices-R team at AI4Bharat for making their Maithili partition available, and the five anonymous listeners who provided the MOS-prosody ratings used in Table II. All remaining errors are the author's own.

REFERENCES

[1] A. Radford *et al.*, “Robust speech recognition via large-scale weak supervision,” in *ICML*, 2023.

[2] A. Gulati *et al.*, “Conformer: Convolution-augmented transformer for speech recognition,” in *Interspeech*, 2020.

[3] J. Gala *et al.*, “IndicTrans2: Towards high-quality and accessible machine translation models for all 22 scheduled Indian languages,” *TMLR*, 2023.

[4] V. Pratap *et al.*, “Scaling speech technology to 1000+ languages,” *JMLR*, 2024.

[5] B. Desplanques, J. Thienpondt, and K. Demuynck, “ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN-based speaker verification,” in *Interspeech*, 2020.

[6] H. Sakoe and S. Chiba, “Dynamic programming algorithm optimization for spoken word recognition,” *IEEE Trans. ASSP*, vol. 26, 1978.

[7] X. Wu *et al.*, “A light CNN for deep face representation with noisy labels,” *IEEE TIFS*, vol. 13, 2018.

[8] I. Goodfellow, J. Shlens, and C. Szegedy, “Explaining and harnessing adversarial examples,” in *ICLR*, 2015.

[9] S. Kneser and H. Ney, “Improved backing-off for m-gram language modeling,” in *ICASSP*, 1995.

[10] K. Prahallad *et al.*, “IndicVoices-R: Unlocking a massive multilingual multi-speaker speech corpus for scaling Indian TTS,” in *NeurIPS Datasets & Benchmarks*, 2024.

[11] T. Javed *et al.*, “Towards building ASR systems for the next billion users,” in *AAAI*, 2022. [Kathbath corpus]

[12] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, “LibriSpeech: An ASR corpus based on public domain audio books,” in *ICASSP*, 2015.

[13] K. Heafield, “KenLM: Faster and smaller language model queries,” in *WMT*, 2011.

[14] EBU Tech 3341, “Loudness metering for broadcasting,” EBU R128, 2016.

[15] C. Gulcehre *et al.*, “On using monolingual corpora in neural machine translation,” arXiv:1503.03535, 2015.

[16] E. McDermott, H. Sak, and E. Variani, “A density ratio approach to language model fusion in end-to-end ASR,” in *ASRU*, 2019.

[17] E. Casanova *et al.*, “YourTTS: Towards zero-shot multi-speaker TTS and zero-shot voice conversion for everyone,” in *ICML*, 2022.

[18] J. Kim, J. Kong, and J. Son, “Conditional variational autoencoder with adversarial learning for end-to-end text-to-speech (VITS),” in *ICML*, 2021.

[19] D. Snyder *et al.*, “X-vectors: Robust DNN embeddings for speaker recognition,” in *ICASSP*, 2018.

[20] M. Ravanelli *et al.*, “SpeechBrain: A general-purpose speech toolkit,” arXiv:2106.04624, 2021.

[21] NIST, “The 2008 NIST language recognition evaluation plan,” 2008.

[22] G. Hinton, “Products of experts,” in *ICANN*, 1999.